Validation

Why regularisation, four objectives, one score

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Weekend 1

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Section 1

Why regularisation

Two numbers

$$\boxed{R_{\text{emp}}(w) = \frac{1}{n}\sum_{i=1}^{n} L(y_i, \hat{y}_i)} \xrightarrow{\textit{hope}} \boxed{R(w) = \int L(y, \hat{y})\, dP(x, y)}$$

what you can compute

what you are paid for

Least squares, 203 countries, 203 columns:

0.997 on the data it fitted, −9.41 on the data it had not.

The gap has a price

$$\boxed{R(w)} \leq \boxed{R_{\text{emp}}(w)} + \boxed{\Omega\left(\frac{h}{n}\right)}$$

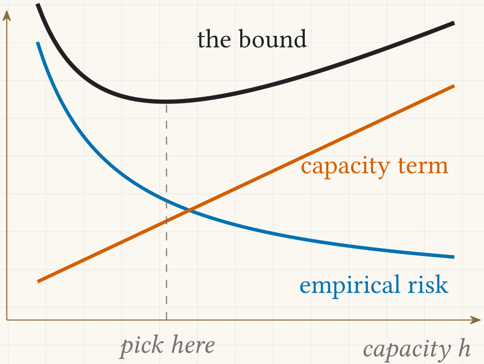
the true risk *what you drove down* *capacity h, sample n*

Zero training error in a large enough family buys nothing.

The second term pays for it.

Structural risk minimisation

- Minimise the sum.
- One α walks the family.
- Large α, small family.



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Section 2

Four objectives

Least squares

$$\boxed{\sum_{i=1}^{n} \left(y_i - \hat{y}_i\right)^2}$$

miss on the data

no penalty

A unique minimiser needs $p < n$.

All 493 columns: 1.000 on train, 0.333 on test.

Ridge

$$\boxed{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2} + \boxed{\alpha \sum_{j=1}^{p} w_j^2}$$

miss on the data

the L2 penalty

Smooth at zero, so nothing lands on zero.

Best test 0.592 at $\alpha = 81.1$, keeping **487 of 493**.

Lasso

$$\boxed{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2} + \boxed{\alpha \sum_{j=1}^{p} |w_j|}$$

miss on the data *the L1 penalty*

A corner at zero, so coefficients land on it.

Best test 0.599 at $\alpha = 285$, keeping **109 of 493**.

Elastic net

$$\boxed{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} + \boxed{\alpha\left(\rho\sum_j |w_j| + \tfrac{1-\rho}{2}\sum_j w_j^2\right)}$$

miss on the data

both, split by ρ

For columns that move together.

Best test 0.592 at $\alpha = 0.811$, keeping **485 of 493**.

The four side by side

Method	Penalty	Zeroes?	Best test	Kept
Least squares	none	no	0.333	493
Ridge	$\alpha \sum w_j^2$	no	0.592	487
Lasso	$\alpha \sum \lvert w_j \rvert$	yes	0.599	109
Elastic net	both, split by ρ	yes	0.592	485

The three α values are **not on one scale**.

Ridge alone carries no $\frac{1}{2n}$. Compare the scores.

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Section 3

Reading the score

R squared

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}$$

- Your miss, over the miss of always predicting $\bar{y}$.
- No floor.